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LAB EXERCISE 17- CREATING SECRETS IN KUBERNETES

CREATING SECRETS IN KUBERNETES IS AN ESSENTIAL PRACTICE FOR MANAGING SENSITIVE INFORMATION SUCH AS PASSWORDS, OAUTH TOKENS, AND SSH KEYS. LET'S GO THROUGH A LAB EXERCISE THAT DEMONSTRATES HOW TO CREATE DIFFERENT TYPES OF SECRETS IN KUBERNETES.

STEP 1: CREATE A GENERIC SECRET

CREATE A SECRET FROM LITERAL VALUES:

```
KUBECTL CREATE SECRET GENERIC MY-SECRET --FROM-LITERAL=USERNAME=ADMIN --FROM-
LITERAL=PASSWORD='P@SSWORD'
rom-literal=username=admin --from-literal=password='P@ssw0rd'
secret/my-secret created
```

VERIFY THE SECRET:

```
KUBECTL GET SECRETS MY-SECRET -O YAML
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-03-16T04:39:19Z"
  name: my-secret
  namespace: default
  resourceVersion: "11323"
  uid: 7f5163f0-6c7e-48e5-8acb-244f1b457dc1
type: Opaque
```

THE OUTPUT WILL BE BASE64 ENCODED. TO DECODE, YOU CAN USE:

```
ECHO -N "P@SSWORD" | BASE64
```

STEP 2: CREATE A SECRET FROM A FILE

CREATE A FILE WITH YOUR SENSITIVE DATA:

```
ECHO -N 'MY-USERNAME' > ./USERNAME.TXT
ECHO -N 'MY-PASSWORD' > ./PASSWORD.TXT
```

CREATE A SECRET FROM THE FILE:

```
KUBECTL CREATE SECRET GENERIC MY-FILE-SECRET-2 --FROM-FILE=USERNAME=./USERNAME.TXT --FROM-
FILE=PASSWORD=./PASSWORD.TXT
```

```
t --from-file=username=./username.txt --from-file=password=./password.txt
secret/my-file-secret created
```

VERIFY THE SECRET:

KUBECTL GET SECRETS MY-FILE-SECRET -O YAML

```
apiVersion: v1
data:
  password: bXktcGFzc3dvcmQ=
  username: bXktcGFzc3dvcmQ=
kind: Secret
metadata:
  creationTimestamp: "2026-03-16T04:40:19Z"
  name: my-file-secret
  namespace: default
  resourceVersion: "11401"
  uid: 89860d6d-d8c0-41b1-8562-db304e6bec5a
type: Opaque
```

STEP 3: CREATE A DOCKER REGISTRY SECRET

CREATE A DOCKER REGISTRY SECRET:

KUBECTL CREATE SECRET DOCKER-REGISTRY MY-REGISTRY-SECRET --DOCKER-USERNAME=<YOUR-USERNAME> --DOCKER-PASSWORD=<YOUR-PASSWORD> --DOCKER-EMAIL=<YOUR-EMAIL> --DOCKER-SERVER=<YOUR-DOCKER-SERVER>

```
gistry-secret --docker-username=sakshamjoshii --docker-password= --docker-email=saksham2006joshi@gmail.com --docker-server=https://index.docke
r.io/v1/
```

VERIFY THE SECRET:

KUBECTL GET SECRETS MY-REGISTRY-SECRET -O YAML

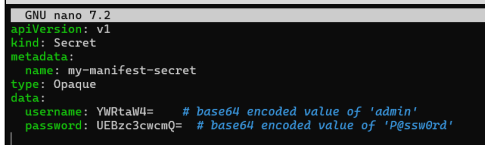
```
aml
apiVersion: v1
data:
  .dockerconfigjson: eyJhdXRob3I6eyJodHRwczovL2luZGV4LmRvY2tldCI5pbY92MS8iOnsidXNlc5hbWUiOiJzYWtzaGFtam9zaGlwIiwicGFzc3dvcmQiOiJDb21wcmVzc29yQDk4MSIsImVtYWlsIjoic2Frc2hhbXcyMDA2am9zaGlAZ21haWwvY29tIiwieXV0aCI6ImMyRnJjMmhoYVldwmdMyaHBhVHBEYjIxd2NtVnpjMjlsUURzNE1RPt0ifX19
kind: Secret
metadata:
  creationTimestamp: "2026-03-16T04:48:31Z"
  name: my-registry-secret
  namespace: default
  resourceVersion: "11774"
  uid: 276cc972-5529-4162-b571-4528bdfd7b6a
type: kubernetes.io/dockerconfigjson
```

STEP 4: CREATE A SECRET USING A YAML MANIFEST

CREATE A YAML FILE FOR THE SECRET:

CREATE A FILE NAMED **MY-SECRET.YAML** WITH THE FOLLOWING CONTENT:

```
APIVERSION: v1
KIND: SECRET
METADATA:
  NAME: MY-MANIFEST-SECRET
TYPE: OPAQUE
DATA:
  USERNAME: YWRtaW4= # BASE64 ENCODED VALUE OF 'ADMIN'
  PASSWORD: UEBzc3cwcmQ= # BASE64 ENCODED VALUE OF 'P@SSWORD'
```



```
GNU nano 7.2
apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
type: Opaque
data:
  username: YWRtaW4= # base64 encoded value of 'admin'
  password: UEBzc3cwcmQ= # base64 encoded value of 'P@ssw0rd'
```

APPLY THE YAML FILE:

```
KUBECTL APPLY -F MY-MANIFEST-SECRET.YAML
```

VERIFY THE SECRET:

```
KUBECTL GET SECRETS MY-MANIFEST-SECRET -O YAML
```

```
aml
apiVersion: v1
data:
  password: UEBzc3cwcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"v1","data":{"password":"UEBzc3cwcmQ=","username":"YWRtaW4="},"kind":"Secret","metadata":{"annotations":{},"name":"my-manifest-secret"},"n
  namespace: "default", "type": "Opaque"}
  creationTimestamp: "2026-03-16T04:50:08Z"
  name: my-manifest-secret
  namespace: default
  resourceVersion: "11851"
  uid: 613f5054-857f-4c52-bd5d-e261e0f0639b
type: Opaque
```

STEP 5: USE SECRETS IN PODS

CREATE A POD THAT USES THE SECRET:

CREATE A FILE NAMED **POD-WITH-SECRET.YAML** WITH THE FOLLOWING CONTENT:

```
APIVERSION: v1
KIND: POD
METADATA:
  NAME: POD-WITH-SECRET
```

SPEC:

CONTAINERS:

- NAME: MYCONTAINER

IMAGE: NGINX

ENV:

- NAME: USERNAME

VALUEFROM:

SECRETKEYREF:

NAME: MY-SECRET

KEY: USERNAME

- NAME: PASSWORD

VALUEFROM:

SECRETKEYREF:

NAME: MY-SECRET

KEY: PASSWORD

APPLY THE YAML FILE:

```
KUBECTL APPLY -F POD-WITH-SECRET.YAML
```

VERIFY THE POD IS RUNNING AND THE SECRET IS BEING USED:

```
KUBECTL EXEC POD-WITH-SECRET -- PRINTENV | GREP -E 'USERNAME|PASSWORD'
```

```
grep -E 'USERNAME|PASSWORD'  
USERNAME=admin  
PASSWORD=P@ssw0rd
```

STEP 7: CLEAN UP

DELETE THE SECRETS:

```
KUBECTL DELETE SECRET MY-SECRET
```

```
KUBECTL DELETE SECRET MY-FILE-SECRET
```

```
KUBECTL DELETE SECRET MY-REGISTRY-SECRET
```

```
KUBECTL DELETE SECRET MY-MANIFEST-SECRET
```

DELETE THE POD:

```
KUBECTL DELETE POD POD-WITH-SECRET
```